

EXPERIMENT 4 — MONITOR, CAPTURE AND ANALYSE NETWORK PACKETS USING WIRESHARK

Aim

To monitor, capture and analyse network packets using Wireshark and understand different network protocols.

Description

Wireshark is an open-source network packet analyser. It captures packets through a selected network interface and provides detailed information for troubleshooting, protocol analysis and security analysis. The captured packets can be inspected using packet lists, protocol details and display filters.

Requirements

Platform	Requirements
Windows	Windows computer, Wireshark, required packet-capture component such as Npcap, network connection
Kali Linux	Kali Linux, Wireshark, Terminal, network connection

PROCEDURE

A. WINDOWS PROCEDURE

Step 1 — Install Wireshark

Install Wireshark on the Windows system. Install the required packet-capture component such as Npcap when prompted.

Step 2 — Launch Wireshark

Open Wireshark from the Start menu or application launcher.

Step 3 — Select the network interface

Select the network interface through which traffic is being transmitted, such as the active Ethernet or Wi-Fi interface.

Step 4 — Start packet capture

Start the capture on the selected interface. Packets travelling through the interface will begin appearing in the packet list.

Step 5 — Generate network traffic

Generate or observe normal network traffic so that packets are captured.

Step 6 — Stop the capture

Stop the capture after sufficient packets have been obtained.

Step 7 — Filter and analyse

Use display filters to focus on particular protocols, hosts or traffic types.

Examples of Display Filters

```
tcp
```

```
dns
```

```
ip.addr == 192.168.1.1
```

B. KALI LINUX PROCEDURE

Step 1 — Update package information

```
sudo apt update
```

Step 2 — Install Wireshark

```
sudo apt install wireshark
```

Step 3 — Launch Wireshark

```
wireshark
```

Step 4 — Select the required interface

Select the active network interface and start packet capture.

Step 5 — Generate and capture traffic

Allow network traffic to pass through the selected interface and observe the captured packets.

Step 6 — Stop and analyse

Stop the capture after sufficient packets have been obtained and inspect individual packets using the packet list and packet-details sections.

Command-line analysis using TShark

TShark is the command-line packet-analysis utility associated with Wireshark.

```
sudo tshark -i eth0
```

```
sudo tshark -i wlan0
```

Packet Information Analysed

- Source address
- Destination address
- Protocol
- Source port
- Destination port
- Packet length
- TCP flags

Capture Filter and Display Filter

A capture filter limits what is captured, while a display filter controls what is shown from an existing capture. Display filters can therefore be used to focus the analysis on particular protocols or hosts after packets have already been captured.

RESULT ANALYSIS

Network packets were successfully captured through the selected interface. The packets contained information such as source address, destination address, protocol, source port, destination port, packet length and TCP flags. Filtering made it possible to focus on particular protocols or hosts. For example, the tcp display filter displays TCP packets, while an IP filter can display traffic associated with a particular address.

Wireshark provides visibility into network communication by allowing individual packets to be captured and examined. This makes it useful for network troubleshooting and security analysis.

RESULT

Network packets were successfully captured, filtered and analysed using Wireshark.

OBSERVATIONS

1. Wireshark captures network packets.
2. A network interface must be selected before capture.

3. Source and destination addresses can be observed.
4. Different protocols can be identified.
5. TCP and UDP packets can be filtered.
6. IP addresses can be used as filters.
7. Ports can be used as filters.
8. Captures can be saved.
9. TShark can be used from the Kali terminal.
10. Wireshark is useful for network troubleshooting and security analysis.

Practical Flow: Install Wireshark → Select network interface → Start capture → Generate network traffic → Stop capture → Apply filters → Inspect packet details → Save the capture if required.